Town Lighting

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Overview

We're adding lighting to Town! This is leveraging Unity's built in lighting system (Unity - Manual: Types of Light component) to add a cozy feeling to town and visual variety over the course of play.

Adding Lights to Decos/Buildings/Etc

There are several ways to approach "lights"; which approach you use depends on the use case and desired look. This guide should help you (the artist) decide which type to use, but whenever in doubt feel free to consult engineering.

Squid Light

When to Use

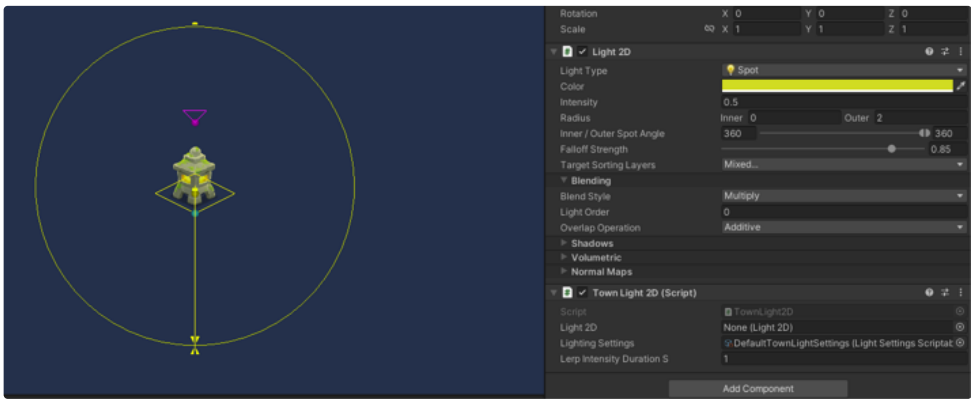
- A SquidLight is our custom wrapper for the default Unity light source, so best to familiarize yourself with Unity lights
- A SquidLight is useful when the light is **meant to illuminate other objects** around the deco, such as a lamp post or fire
- Keep in mind SquidLights are the most expensive version of lighting, so please use sparingly. You shouldn't need more than 1-2 max on a deco

Note the light brightens the deco, the terrain, and nearby decos and characters

How to Use

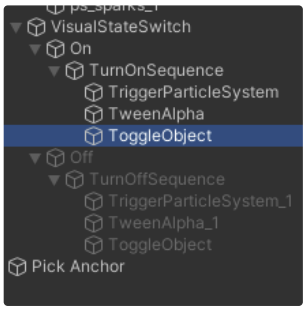
- Create a new GameObject under the AnimRoot and call it something like “Light”
- Add a **TownLight2D** component
  - This is a custom script that will adjust the brightness throughout the day to not blow out brightness during peak daylight but appear bright at night
  - We use 2D spot lights by default, which will always shine on top of everything else in the scene (due to how Unity processes lights). Keep this in mind when setting brightness and position to avoid weird looking overlaps.
- TownLight2D will apply a series of default settings which are defined in `DefaultTownLightSettings.asset`

> Take some time as you work with several decos to define a good standard default, this should help save us time hand tuning lights for every single lit deco



- Adjust the position and light settings on the Light until it looks nice
- Most lit decos have an existing VisualStateSwitch or UISequencer to toggle lights on and off. Add a hook to enable and disable your new Light here as well

- eg This deco has a sequencer that enables particles and alpha fades in a glow sprite; I've added a ToggleObject here to simply toggle the Light on at the same time (and same for the off sequence)



- Test your changes in Unity (see testing section below)

Emissive Sprite

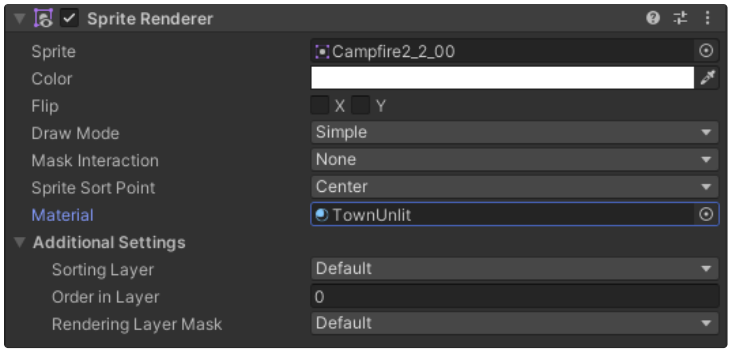
When to Use

- The tl;dr for lights is as Unity draws the scene it will apply a multiplier to darken or brighten each pixel based on how many lights are near it (this can get very expensive very quickly)
- Often times it's much faster and just as good to fake lights by adding sprites that simply ignore this light calculation. **This creates an illusion of the sprite emitting its own light** (we call this an emissive property).
- This is useful for things that should self-glow like neon signs or a flame, where it's less important for nearby objects to receive that light
  - Campfire is going to be a bit of a weird example because we want the flame sprite to glow itself without being affected by lights, but also some amount of light to illuminate nearby objects. We can use both of these techniques together to get the desired look.

The flame sprite is ignoring lighting (the true light that glows around it has been disabled for this example)

How to Use

- Look for the Sprite Renderer component in the prefab that you want to be emissive.



- Locate where the Material is set to `TownLit` and swap it out for `TownUnlit`
- That's it! Now the sprite will ignore lighting calculations entirely and just be rendered at full brightness

Spine Lighting

- For spine animated decos we can export them with multiple materials to separate out the lit and unlit or additive pieces. Ask Ivan to author spine decos with lighting in mind so he can export the correct materials (see Flamingod as an example deco with emissive glowing pieces but a light-respecting body)

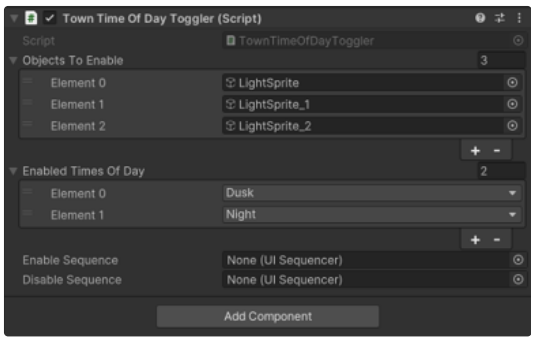
Custom Shader

- This should be used very sparingly and for things that already use their own custom shading (eg water, fog, etc). Ask Caroline if something that isn't a deco/building looks incorrectly lit; most shaders have been migrated to lit versions already.

Auto Toggling Lights

- Most lit decos currently let the user toggle the lights on and off at will, but we may want some lights to turn on and off at certain times of day (such as windows in homes or an open/close sign on a business). We have a component that can be set up easily to allow these togglings.
- Open the prefab and create a GameObject root to hold your togglables (or just drive the logic). Add a `TownTimeOfDayToggler` component and hook up your togglable objects to "Objects to Enable." Set up the times of day you want the objects to be on for (will be off for all other times)

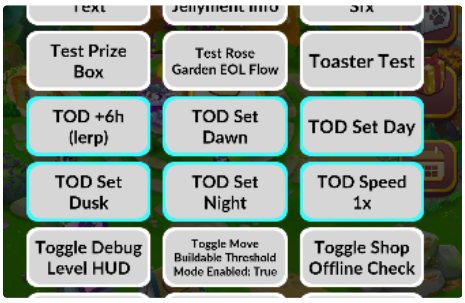
In this example the window sprites are set to be emissive (unlit material)



We support sequencers too if you want the lights to fade in/out instead of just snapping on and off

Migrating Existing Prefabs to Lit Variants

- As we spin up the lighting branch there are decos and buildings being added that are still using the old unlit material (or maybe a new deco was added and the person setting it up forgot to use the lit material). Luckily we have tooling in place to help with this.
- Simply right click on the prefab and select `Migrate to Lit Material`. This will swap all Sprite Renderers to use the TownLit material and all Spine materials to use the Skeleton Lit shader.
  - Note for Spine assets the tool may fail if there are a combination of different materials applied, please ask Ivan or an engineer to adjust the materials manually. The shader we use for lit Spine skeletons is `Starfish/Universal Render Pipeline/2D/Spine/Skeleton Lit`
- From there edit the prefab as normal to add lights and unlit emissive sprites

Testing Lighting in Town

- Our Time of Day system is primarily tuned via `TimeOfDaySettings.asset`. This controls the time window for each phase of the day (Dawn, Day, Dusk, Night), the ambient light color for town, and the duration of the in-game day.
- To test the look of your new lit deco, add it to town using the debug inventory tool to grant an instance to yourself, and place it in town.
- You can use the debug menu TOD buttons to jump between times of day